USE CASE FLOW
UC-01 — Raise Emergency Blood Request

Preconditions

- The emergency requester has access to the blood bank system.
- The blood bank inventory is available for checking.

Postconditions

- The emergency blood request is recorded with a unique request ID and current status.
- If sufficient blood is unavailable, compatible eligible donors are identified and emergency notifications are sent.

Main Success Scenario

1. Emergency Requester selects “Raise Emergency Blood Request.”
2. The system displays the emergency request form.
3. The requester enters the required blood group, component, quantity, and location.
4. The system validates the emergency request.
5. The system checks the real-time blood bank inventory.
6. The system determines that the required blood is available.
7. The system records the emergency blood request.
8. The system displays confirmation and the current request status.
9. The use case ends successfully.

Alternate Flow — Insufficient Blood Stock

1. At Step 6, the system determines that the available blood is insufficient.
2. The system initiates donor matching for the requested blood group.
3. The system identifies compatible and eligible donors within a 10 km radius.
4. The system sends emergency SMS notifications to the matching donors.
5. The system marks the emergency request as awaiting donor response.

UML relationship reflected: UC-01 includes “Check Blood Inventory”; donor matching extends the emergency request when stock is insufficient.